

HABIB UL REHMAN

01

9 HIGH-IMPACT QUICK WINS



ASP.NET CORE WEB API



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Version Early

Add Asp.Versioning.* now. Breaking change without versioning = silent client pain + rushed hotfixes.

```
1 services.AddApiVersioning(options =>
2 {
3     options.DefaultApiVersion = new ApiVersion(1, 0);
4     options.AssumeDefaultVersionWhenUnspecified = true;
5     options.ReportApiVersions = true;
6 });
```

Shape Contracts With DTOs

Never return EF entities. Explicit DTOs let you evolve fields safely and hide persistence leaks.

```
public class ProductDto
{
    public int Id { get; set; }
    public string Name { get; set; }
}
```

Centralize Validation

Kill scattered if (string.IsNullOrEmpty(...)). Use FluentValidation + automatic 400 with ProblemDetails.

```
// Install FluentValidation and register validators
services.AddFluentValidationAutoValidation();
services.AddValidatorsFromAssemblyContaining<Program>();

public class ProductDtoValidator : AbstractValidator<ProductDto>
{
    public ProductDtoValidator()
    {
        RuleFor(x => x.Name).NotEmpty();
    }
}
```

Consistent Errors

One middleware to turn exceptions into typed ProblemDetails. No stack traces to clients, rich logs internally.

```
app.UseExceptionHandler("/error");

app.Map("/error", (HttpContext httpContext) =>
{
    var feature = httpContext.Features.Get<IExceptionHandlerFeature>();
    return Results.Problem(detail: feature?.Error.Message);
});
```

Caching Strategy (2 Layers)

- a) Response Caching / CDN for idempotent GETs.
- b) Memory / distributed cache (keyed by version + query).
Invalidate on writes—don't guess.

```
[ResponseCache(Duration = 60)]  
[HttpGet]  
public IActionResult GetProducts() => Ok(_service.GetProducts());  
  
// In-memory cache example  
var products = memoryCache.GetOrCreate("products_v1", entry =>  
{  
    entry.AbsoluteExpirationRelativeToNow = TimeSpan.FromMinutes(10);  
    return _service.GetProducts();  
});
```

Observability As Code

Add OpenTelemetry (traces + metrics) + structured logging (CorrelationId). You can't tune what you can't see.

```
// Structured logging
logger.LogInformation("Getting product with id {ProductId}", id);

// Enable OpenTelemetry in Program.cs
services.AddOpenTelemetryTracing(builder =>
    builder.AddAspNetCoreInstrumentation().AddConsoleExporter());
```

Cancellation & Timeouts

Accept CancellationToken on every async endpoint + outbound HTTP calls. Fewer zombie tasks, faster resource recovery.

```
[HttpGet]
public async Task<IActionResult> GetProducts(CancellationToken cancellationToken)
{
    var products = await _service.GetAsync(cancellationToken);
    return Ok(products);
}
```


Use Attribute Routing

Employ attribute routing for Web API controllers to define clear and concise endpoint routes.

```
1 [Route("api/[controller]")]
2 public class ProductsController : ControllerBase
3 {
4     [HttpGet("{id}")]
5     public IActionResult GetProduct(int id)
6     {
7         // Get product logic
8     }
9 }
```

Use Dependency Injection for Services

Leverage Dependency Injection to inject services into controllers, promoting modularity and testability.

```
1 public class ProductsController : ControllerBase
2 {
3     private readonly IProductService _productService;
4
5     public ProductsController(IProductService productService)
6     {
7         _productService = productService;
8     }
9 }
```

✓ **ACTION TODAY: PICK ONE OF THE 9 AND IMPLEMENT IT BEFORE LUNCH. MOMENTUM COMPOUNDS.**

☞ **WHICH ONE IS MISSING IN YOUR CURRENT API—AND WHY HASN'T IT BEEN ADDED YET?**



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